

A Vocabulary of Axes Generated by an Inverse Doubling, and the Parity Construction It Supports

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Abstract

We describe a finite construction of straight-line figures on a square, and we establish that its vocabulary is **generated** rather than chosen.

One operation suffices, and the origin is forced rather than chosen. The operation is the doubling of the offset about the centre of the square; the vocabulary is the smallest set of lines stable under the inverse of that doubling. A level fixed by the doubling has no preimage outside itself, so it cannot be generated and must be posited, while every other level is reachable: the origin is therefore the set of fixed levels, and there are exactly two of them, 0° and 120° , the second existing precisely because 3 divides 12. Started from those two, the orthogonal vocabulary closes in two steps on 24 offsets, exactly the union of the eight orthogonal families; the diagonal vocabulary closes in three steps, also on 24, the union of the eight diagonal families together with one further offset — the inscribed diamond, which belongs to the vocabulary and is not drawn.

The first generation of that closure — four families — is not an arbitrary stopping point: its fifteen non-empty unions are, one for one, the fifteen families of a pattern system fixed by hand years earlier, and the check against the sixty published images succeeds sixty times out of sixty.

Three consequences follow, none of them stipulated. The backward structure is not one chain but two towers of height two, rooted at the two fixed levels, and at one place only — the top of the first tower — the two branches of the inverse doubling coincide with the division of a level into a *base* and a *mutant*; what holds at every level is the shift $e \mapsto e + 3 \pmod{6}$, verified on all four, and it is that shift which defines the mutation. The doubling has exactly four fixed families, and an explicit criterion — verified on all 65 535 subsets without exception — decides when iterating it leaves the vocabulary. Finally, the parity rule that turns a set of axes into a figure is reversible: it is injective on band systems outright, and on the plates tested a figure determines the chord that produced it up to the three relations the families satisfy.

We also show that the cell on which the parity is read is a parameter and not a constant: the same axes and the same rule, read on a triangle subdivision or on the square cell, produce two figures that share no information, and we give the scalar that says which reading carries content. The integrity check the construction affords is a linear code: the sixteen family figures span $[1152, 13]$ with minimum distance 288 on the triangle reading and $[144, 12]$ with distance 36 on the cell reading, so the corpus corrects errors and does not merely detect them. And the period 12 is itself constrained: the descent terminates without a cycle exactly for $N = 2^a$ or $3 \cdot 2^a$, and a non-trivial fixed level exists exactly when $3 \mid N$; requiring in addition a depth of at least two leaves 12, 24, 48.

Version 1.1 adds two things. The cell reading C_1 now states its rule for axes that pass through cell centres — such an axis does not separate the cell, which closes the figure continuously — and every C_1 number depends on it (§5.4). And a **second cut** of the diagonal vocabulary is examined (§5.5): twelve elements, YA1–YA6 and AY1–AY6, which partition the 288 cell diagonals of the square and which, read as band systems, give exactly **one generator per node of the doubling tree**. The first cut resolves one tower node by node and the other only by pairs; the second resolves both. The code passes from $[1152, 13, 288]$ to $[1152, 15, 288]$ on C_8 and from $[144, 12, 36]$ to $[144, 14, 36]$ on C_1 ; it contains the old code, keeps the same minimum-weight words, and has no relation among its diagonal generators. On the second cut the mutation is the shift $e \mapsto e + 3$ on the diagonals as well as on the orthogonal levels.

This is a working basis and not a finished result. We claim no depth for the construction and make no physical claim; the comparison with the nodal and iso-amplitude figures of a vibrating plate is the subject of a separate paper, which uses the vocabulary established here and nothing else from it.

1. The square, the natures, the offset

Let P be the square $[0, 12]^2$ divided into 144 unit cells.

An **axis** is a straight line drawn on P in one of four **natures** — horizontal, vertical, and the two diagonal directions. It is specified by its signed **offset** e from the centre, measured in cells; for the diagonal natures the offset is the perpendicular coordinate rescaled to cells, that is $(x + y - 12)/2$ and $(y - x)/2$. Offsets lie in $(-6, 6]$, with 6 and -6 denoting the same line, and two axes of the same nature and different offsets are different lines.

Two subdivisions of P are used. C_1 takes each cell as one region, giving 144 regions. The test point of a region and the rule for an axis passing through it are fixed in §5.4. C_8 cuts each cell by its two diagonals and its two medians into eight triangles, giving 1152 regions;

a region is indexed by (r, c, s) with s the 45° sector clockwise from the upward vertical. The sectors are not angularly uniform — the centroids fall at $\pm 26.57^\circ$ and $\pm 63.43^\circ$ — so a sector must be assigned by $s = \lfloor \theta/45 \rfloor$ and not by rounding to the nearest multiple.

A **figure** is a two-colouring of the regions of one subdivision.

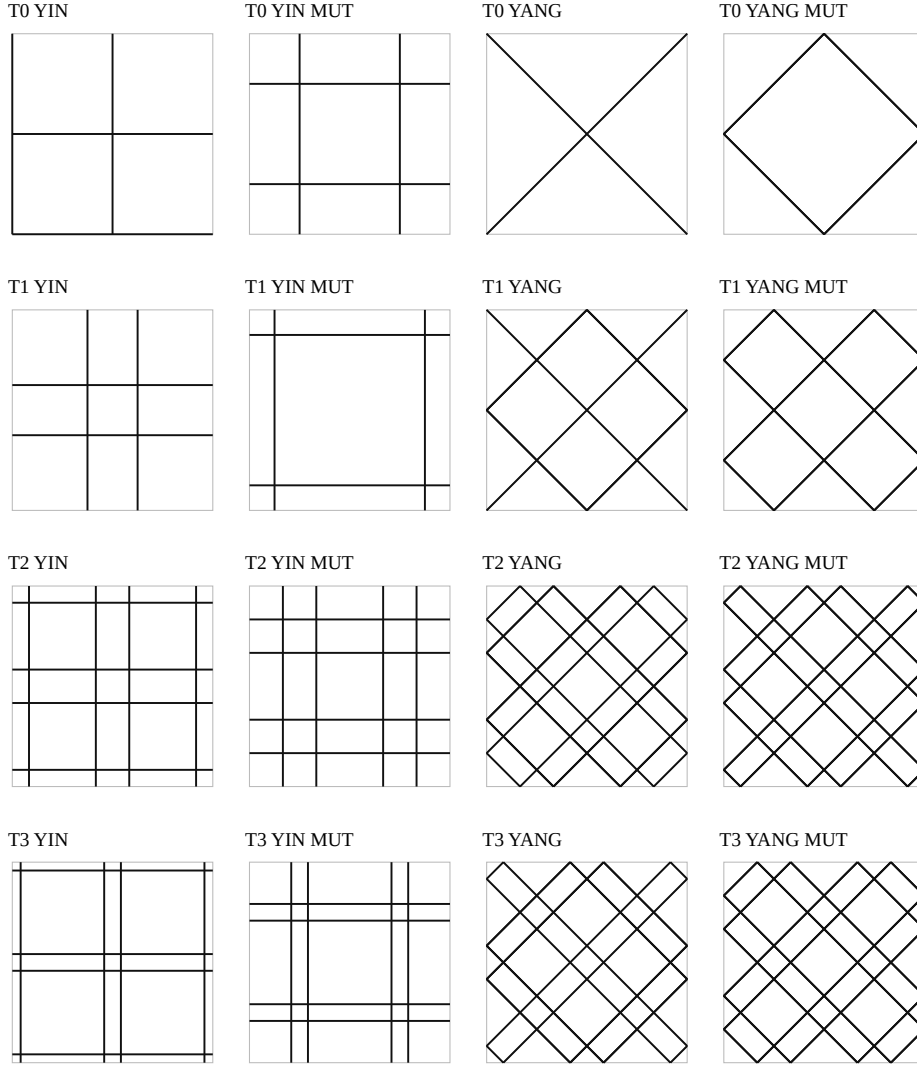


Figure 1. The sixteen families, drawn on the square. The frame is the boundary of P ; an axis lying on it is drawn in full. Counted as segments there are 94 distinct lines here, thirty of them appearing in two families.

2. The vocabulary is generated

Let σ be the doubling of the offset about the centre,

$$\sigma: e \mapsto 2e \pmod{12},$$

acting within each nature. It is the homothety of ratio 2 read on offsets. It is two-to-one: every offset has exactly two preimages, $e/2$ and $e/2 + 6$.

Definition. Given a set Ω of offsets in one nature, the vocabulary generated by Ω is the smallest set containing Ω and stable under σ^{-1} .

Nothing else is stipulated: there is no free parameter in this definition, hence nothing that could have been tuned.

What is to be taken as origin is not free either, and this is the point of the section. A level fixed by σ has no preimage outside itself, so it cannot be reached by any number of backward steps: it must be posited. Conversely, every level that is *not* fixed is reachable. The origin is therefore forced to be the set of fixed levels, and Theorem 8 of §6 says what that set is: the level 0° always, and the level 120° exactly because 3 divides 12.

In each nature those two levels are two of the sixteen families:

$$\begin{array}{ll} \text{orthogonal} & 0^\circ = T_0 \text{ YIN } \{0, 6\} \quad 120^\circ = T_2 \text{ YIN MUT } \{\pm 2, \pm 4\} \\ \text{diagonal} & 0^\circ = T_0 \text{ YANG } \{0\} \quad 120^\circ = T_2 \text{ YANG } \{\pm 1, \pm 2, \pm 4\} \end{array}$$

Theorem 1 (orthogonal). *The vocabulary generated by the two fixed orthogonal levels, $\Omega_o = \{0, 6\} \cup \{\pm 2, \pm 4\}$, reaches a fixed point in two steps, on 24 offsets. That fixed point is exactly the union of the eight orthogonal families of the system.*

The sizes are 6, 12, 24.

Theorem 2 (diagonal). *The vocabulary generated by the two fixed diagonal levels, $\Omega_d = \{0\} \cup \{\pm 1, \pm 2, \pm 4\}$, reaches a fixed point in three steps, also on 24 offsets. That fixed point is the union of the eight diagonal families together with the single offset 6.*

The sizes are 7, 14, 20, 24.

Neither level generates the other, and neither suffices alone. The 0° level closes, in *either* nature, on the same eight offsets $\{0, \pm 1.5, \pm 3, \pm 4.5, 6\}$ — four families orthogonally, three plus the diamond diagonally — and stops there, because ± 1.5 has no preimage on the half-integer grid. The 120° level closes on sixteen offsets orthogonally and never produces ± 1.5 at all. The two towers are disjoint above their roots and their union is the whole vocabulary. This is why §2.1 below concerns the 0° tower alone: the fifteen families of the existing system are generated by that root, and the 120° tower is what the plates of the treatise add to it.

The offset 6. In the orthogonal natures it is the edge of P , drawn; in the diagonal natures it is the inscribed diamond, in the vocabulary but *not* drawn. In both it is the second preimage of 0, the one that the fixed level does not already contain, and it is what lets the 0° tower rise at all. Forbidding it at every step costs exactly the same seven offsets in both natures — $\{\pm 1.5, \pm 3, \pm 4.5, 6\}$, the whole tower above its root — and leaves 17 offsets instead of 24. Orthogonally that is T_0 YIN MUT, T_1 YIN and T_1 YIN MUT; diagonally, T_0 YANG MUT and T_1 YANG MUT. The two natures have the same vocabulary of 24 offsets, the same origin structure and the same operation, and differ only in the status of that one offset — drawn on one side, an invisible generator on the other.

2.1 The first generation is the vocabulary of an existing system

The first step of the 0° tower orthogonally, and its second step diagonally, produce four families: T_0 YIN, T_0 YIN MUT, T_0 YANG and T_0 YANG MUT. They are pairwise disjoint, so under symmetric difference they generate a space isomorphic to $(\mathbb{Z}/2)^4$, whose fifteen non-zero elements are the fifteen non-empty unions.

Those fifteen unions are not an abstraction. The pattern system from which this vocabulary is drawn was designed by hand and organised, long before any of this, into fifteen families of four *natures* — sixty images in all. Checked against the published data:

- every one of the sixty images has one of its three tints occupying exactly the forty-eight cells whose centre lies on an axis of T_1 YANG, the same set for all sixty (thirty images in one tint, thirty in another);
- the remaining ninety-six cells of every image are, in every case, the parity figure of one of the fifteen unions of T_0 families, split between the two remaining tints.

Sixty out of sixty, by exhaustive comparison. **The fifteen families of the existing system are the fifteen non-empty subsets of the four families of the first generation**, and the four natures are the four tint permutations that distinguish the readings.

This is the one justification we can give for the origin of §2. It is not that $\{0, 6\}$ is canonical among all possible origins; it is that its first generation reproduces, exactly and without residue, a vocabulary that was fixed independently and beforehand.

Counted as drawn lines across the four natures and the sixteen families, the vocabulary has **94 distinct axes**. Thirty of them belong to two families each, and all thirty are diagonal; no orthogonal axis is shared.

3. Two towers, and what separates base from mutant

The backward structure is not a single chain. Written out in full, the orthogonal preimages are

$$\begin{array}{ll} \sigma^{-1}(T_0 \text{ YIN}) & = T_0 \text{ YIN} \cup T_0 \text{ YIN MUT} \\ \sigma^{-1}(T_0 \text{ YIN MUT}) & = T_1 \text{ YIN} \cup T_1 \text{ YIN MUT} \\ \sigma^{-1}(T_1 \text{ YIN}) & = \sigma^{-1}(T_1 \text{ YIN MUT}) = \emptyset \\ \sigma^{-1}(T_2 \text{ YIN MUT}) & = T_2 \text{ YIN} \cup T_2 \text{ YIN MUT} \\ \sigma^{-1}(T_2 \text{ YIN}) & = T_3 \text{ YIN} \cup T_3 \text{ YIN MUT} \\ \sigma^{-1}(T_3 \text{ YIN}) & = \sigma^{-1}(T_3 \text{ YIN MUT}) = \emptyset \end{array}$$

so there is no chain $T_0 \rightarrow T_1 \rightarrow T_2 \rightarrow T_3$. There are two towers of height two, rooted at the two fixed levels of §2:

$$T_0 \text{ YIN} \rightarrow T_0 \text{ YIN MUT} \rightarrow \{T_1 \text{ YIN}, T_1 \text{ YIN MUT}\}, \quad T_2 \text{ YIN MUT} \rightarrow T_2 \text{ YIN} \rightarrow \{T_3 \text{ YIN}, T_3 \text{ YIN MUT}\}.$$

The names $T_0, \dots, T_3$ index levels of φ , not generations of σ , and the two orderings do not agree; we keep the names because the existing system uses them.

Since σ is two-to-one, each preimage set splits into two branches, and at exactly one place in the structure that split coincides with the division into a **base** and a **mutant**: the top of the first tower. The two preimages of 3 are 1.5, in T_1 YIN, and -4.5 , in T_1 YIN MUT. Nowhere else. At the root of that tower the two preimages of 0 are 0 and 6, both in T_0 YIN; at the root of the second, the preimages of 2 are 1 and -5 , both in T_2 YIN; and at the *top* of the second the branches do not separate the families either — the preimages of 1 are 0.5 and -5.5 , both in T_3 YIN, and those of 5 are 2.5 and -3.5 , both in T_3 YIN MUT. So the inverse doubling produces the base/mutant division once, by accident of where the offsets fall, and it is the following that holds throughout.

arrows run in the direction of σ^{-1} ; the descent σ runs against them

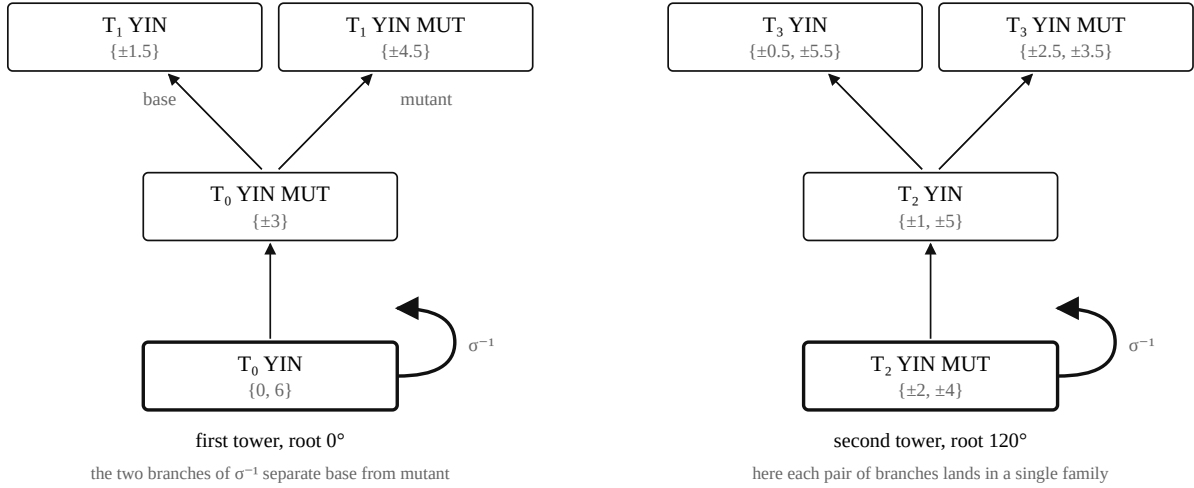


Figure 2. The two orthogonal towers. Boxes are families, arrows are σ^{-1} , and a double border marks a level fixed by σ . The descent of §4 runs against the arrows.

Proposition 1. *On the orthogonal levels the mutant is obtained from the base by $e \mapsto e + 3 \pmod{6}$: for each level t , the offsets of the mutant are those of the base shifted by 3 modulo 6. Verified on all four levels.*

This, and not the inverse doubling, is what defines the mutation. The relation fails on the diagonal families of the first cut, and §2 says why: each of them already carries both branches, so no shift can separate what is not separated. It is a property of the cut and not of the diagonals: on the second cut of §5.5, which carries one level per generator, the same shift relates every yang element to its mutable twin.

4. The descent, and where it leaves the vocabulary

Iterating σ forward instead of backward gives a descent.

Theorem 3. *σ maps each of the sixteen families into another family of the vocabulary, with two exceptions. Writing $A \rightarrow B$ for $\sigma(A) \subseteq B$ with B the smallest family containing $\sigma(A)$, and marking with $\equiv$ the arrows that are equalities,*

$$\begin{array}{ll}
 T_1 \text{ YIN} \xrightarrow{\equiv} T_0 \text{ YIN MUT}, & T_1 \text{ YIN MUT} \xrightarrow{\equiv} T_0 \text{ YIN MUT} \rightarrow T_0 \text{ YIN} \\
 T_3 \text{ YIN}, T_3 \text{ YIN MUT} & \rightarrow T_2 \text{ YIN} \rightarrow T_2 \text{ YIN MUT} \\
 T_1 \text{ YANG MUT} & \xrightarrow{\equiv} T_0 \text{ YANG MUT} \\
 T_2 \text{ YANG MUT}, T_3 \text{ YANG} & \rightarrow T_3 \text{ YANG MUT} \rightarrow T_2 \text{ YANG}
 \end{array}$$

Four arrows are equalities: the three marked above and $T_0 \text{ YANG} \rightarrow T_0 \text{ YANG}$, which is a fixed family. All others are strict inclusions. and the fixed families are exactly

$$T_0 \text{ YIN } \{0, 6\}, \quad T_0 \text{ YANG } \{0\}, \quad T_2 \text{ YIN MUT } \{\pm 2, \pm 4\}, \quad T_2 \text{ YANG } \{\pm 1, \pm 2, \pm 4\}.$$

Theorem 4. *The image of a chord under σ leaves the vocabulary if and only if the chord contains $T_0 \text{ YANG MUT}$ or $T_1 \text{ YANG}$. Verified on all 65 535 chords, with no counterexample.*

The escape is always to the same object, the diagonal offset 6 — the inscribed diamond of §2. The generator of the diagonal vocabulary is also the one place where the descent leaves it.

Two notions of fixity must be kept apart, or the count changes between statements. Theorem 3 calls a family fixed when the arrow returns to it, that is when the *smallest* family containing $\sigma(A)$ is A itself; there are four. The weaker condition $\sigma(A) \subseteq A$ holds for five, the fifth being T_3 YANG MUT, whose image $\{\pm 2, \pm 4\}$ sits inside its own sixteen axes but sits inside the twelve of T_2 YANG as well, so its arrow points at T_2 YANG. The corollary below uses the weaker condition.

Corollary. *Call a chord harmonically closed if $\sigma(A) \subseteq A$. There are 719 such chords, and they are exactly the chords closed downward under the descent; the minimal ones are the five families satisfying that condition — T_0 YIN, T_0 YANG, T_2 YIN MUT, T_2 YANG, T_3 YANG MUT. Harmonic closure is therefore an operation and not a search: adjoin the images of the members until the fixed families are reached.*

5. The parity, and the cell on which it is read

5.1 Drawn lines and stripe systems

A figure is obtained from a chord by a parity rule, and the rule distinguishes two objects that must not be conflated.

The **catalogue** lists drawn lines: offsets in $(-6, 6]$, no folding.

The **parity** operates on periodic stripe systems. Each drawn line is converted to a key $(\text{nature}, u \bmod 12)$, where u is the *raw* perpendicular coordinate — $6 + e$ for a horizontal or vertical axis, $12 + 2e$ for $x + y$, and $2e$ for $y - x$ — and the exclusive-or is taken over the **set** of keys. Because of the factor 2, two diagonal offsets differing by 6 share a key; orthogonal offsets never collide. Seven of the sixteen families therefore have fewer stripe systems than drawn lines: T_2 YANG has twelve lines and eight systems, and those eight are exactly the systems of T_3 YANG MUT, which is why the two families have the same figure and why one of the three relations of §5.4 exists at all.

Deduplication is not cancellation. The parity is an exclusive-or over distinct keys, never over the list of offsets, and a family read alone gives the same figure as the same family read inside a union.

5.2 A convention-free criterion, and reversibility

The definition above fixes an origin and a global polarity, neither intrinsic. The following avoids both.

Definition. A figure F is the *parity* of a set of axes A if, for every pair of adjacent regions, the two differ in colour **if and only if** an axis of A lies along their common edge.

This is differential: insensitive to the origin and invariant under global colour exchange. On C_8 there are 1728 adjacent pairs, 576 of them between cells.

Theorem 5. *On the plates tested, each figure is the parity of a chord of at most six families, recovered by exhaustive sweep over the 14 892 such chords at zero discrepancy over 1152 regions, and that chord is unique modulo the three relations of Theorem 6 below.*

The qualification is not a formality. Under the canonical rule T_2 YANG and T_3 YANG MUT have the same key set and therefore the same figure, as do T_3 YANG and T_2 YANG MUT, so a chord containing one member of such a pair and the chord obtained by swapping it for the other are indistinguishable by their figure, and no amount of sweeping will separate them. What is true without qualification is stated in Theorem 7: the parity map is injective on band systems. Theorem 5 is the restriction of that injectivity to unions of families, where the fibre is the coset of the relation space.

Uniqueness makes the construction a code rather than a drawing method: a figure carries its own generating data, recoverable without external reference.

The generative procedure follows: take a chord of two families and compute the parity; add a third and **recompute** on the union of axes; add a fourth, and so on. It is not a superposition of the previous figures.

5.3 The reading cell is a parameter

Some figures admit no chord on C_8 and are exact on C_1 :

subdivision	pairs tested	figure A	figure B
C_8	1728	288 violations	480 violations
C_1 , between cells	576	0	0

Every violation lies on an edge interior to a cell, none between cells, and they are spread over 120 and 144 of the 144 cells — not a localised defect.

One might expect C_1 to be a coarsening of C_8 . It is not.

Proposition 2. *For those chords, each of the 144 cells contains exactly four dark and four light triangles in the C_8 reading, and the two figures differ on 576 of 1152 regions — exactly half.*

A cell balanced 4/4 carries no cell-level information, so no coarsening can recover the C_1 figure; and half is the maximum possible discrepancy. The two readings are independent channels of the same data: C_8 carries information only inside cells, C_1 only between them.

The number of 4/4-balanced cells measures which reading carries content:

chord	balanced cells
T_0 YIN, T_0 YIN MUT, T_2 YIN, T_2 YIN MUT	0/144
T_0 YANG, T_0 YANG MUT	24/144
T_1 YIN, T_1 YIN MUT	44/144
T_1 YANG, T_1 YANG MUT	48/144
T_3 YIN, T_3 YIN MUT	80/144
T_2 YANG, T_2 YANG MUT, T_3 YANG, T_3 YANG MUT	96/144

No single family reaches 144. The choice of reading cell is therefore not a matter of taste: at 0/144 the triangle reading carries everything, at 144/144 it carries nothing at cell level, and the scalar interpolates.

5.4 The integrity check is a linear code

The consistency property is not merely that each figure is the exclusive-or of two others. The figures of the sixteen families span a linear code over $\mathbb{F}_2$, and its parameters are computable.

The rule for axes through cell centres. On C_8 the test point of a region is its centroid, chosen so that it lies on no axis. On C_1 the test point is the cell centre $(c + \frac{1}{2}, r + \frac{1}{2})$, where $x + y$ and $y - x$ are integers, so that every diagonal key and every orthogonal key at a half-integer offset — 48 of the 72 — passes through cell centres. Such an axis is treated apart: it does not separate the cell, which closes the figure *continuously*, on the side of increasing coordinate, the same all along the axis. In `regle_parity.py` this is the floor in $\lfloor (u - c) / 12 \rfloor \bmod 2$; the reference-based count of `codes_cles.py` places such cells on the same side, and the two implementations agree cell by cell on all sixteen families. The rule is part of the definition of C_1 and every C_1 number below depends on it: a rule that sent such a cell to the side of the reference region instead would lose the all-dark word.

Theorem 6. *On C_8 the sixteen family figures span a code $[1152, 13]$ whose minimum distance is 288, attained by a single word, and these parameters do not depend on the parity convention. On C_1 the minimum distance is 36 and the dimension is 12 if figures carry the polarity the rule gives them, 11 if a reference region is required to be light.*

A minimum distance of 288 detects up to 287 wrong regions and corrects up to 143; at exactly 288 there is a hole, since a corruption of that size can land on another codeword. The unique minimum-weight word is T_0 YANG MUT $\oplus$ T_1 YIN — the diamond against the inner square of offsets ± 1.5 . Naming it requires naming the reference region from which the parity is counted, here the corner region at the origin: with the central region as reference every generator that is dark there is complemented, the code changes, and the minimum-weight word becomes T_0 YANG MUT $\oplus$ T_1 YIN MUT. The distance 288 and the uniqueness do not depend on that choice; the name does. The kernel has dimension 3 at C_8 , generated by

T_0 YANG + T_0 YANG MUT + T_1 YANG = 0, T_2 YANG MUT + T_3 YANG = 0, T_2 YANG + T_3 YANG MUT = 0, the first being the parity translation of the set identity T_1 YANG = T_0 YANG $\cup$ T_0 YANG MUT. At C_1 two further facts hold, and they are distinct. First, a fourth relation appears: the eight YIN families sum to zero. Their 48 keys are pairwise disjoint, so this is a relation in the union semantics as much as in the symmetric difference, and not an artefact of one of the two. Second, and separately, the all-dark figure lies in the span on C_1 and not on C_8 ; its shortest expression is T_1 YANG $\oplus$ T_1 YANG MUT $\oplus$ T_2 YANG $\oplus$ T_2 YANG MUT. So a global inversion is already a combination of the families at cell level, which it is not at triangle level. Read through §5.5, those four families hold the seven diagonal levels between them, each once: the all-dark figure on C_1 is the sum of all the diagonal levels.

The dimension 13 is not only computed; it is read off the levels. Across the diagonals the level of an axis of offset e is $\cos(\pi e/3)$, well defined modulo the period 6, and the eight diagonal families occupy five distinct sets of levels — 0° (T_0 YANG), 180° (T_0 YANG MUT), 90° (T_1 YANG MUT), $\{60^\circ, 120^\circ\}$ (T_2 YANG and T_3 YANG MUT) and $\{30^\circ, 150^\circ\}$ (T_2 YANG MUT and T_3 YANG), with T_1 YANG the union of the first two. So the diagonal families have rank 5, the eight orthogonal ones rank 8 on C_8 , and $5 + 8 = 13$; the three relations are the three diagonal families that repeat what is already spanned. On C_1 the orthogonal rank drops to 7 — the fourth relation — and $5 + 7 = 12$.

That second fact is what makes the dimension at C_1 depend on the convention, and it is worth stating rather than hiding behind a choice. Requiring a reference region to be light quotients the code by the all-dark vector; on C_8 that vector is not in the span and nothing changes, while on C_1 it is, and exactly one dimension is lost — the relation $T_1 \text{ YANG} \oplus T_1 \text{ YANG MUT} \oplus T_2 \text{ YANG} \oplus T_2 \text{ YANG MUT}$, which equals the all-dark figure without the anchoring and zero with it. The eight-YIN relation survives both, since all eight are light at the reference region. The minimum distance is 36 either way.

Theorem 7. *The 94 drawn axes give 72 distinct stripe systems, of which exactly two have an empty mask on C_8 — the two border systems at $u \equiv 0$. The remaining 70 are linearly independent.*

So a figure determines its set of stripe systems uniquely, up to the two invisible border systems, and this holds for an arbitrary set of axes and not only for a chord. A count sometimes quoted alongside these is 18431; it is the number of distinct unions of drawn segments over the 65 535 chords, not of figures. The figures are $8191 = 2^{13} - 1$ under the union semantics, which never produces the empty figure on C_8 , and 8192 words under symmetric difference, seven chords — of sizes 2, 2, 3, 4, 5, 5, 7 — giving the empty one. Theorem 5 is the restriction of that injectivity to unions of families.

5.5 A second cut: one generator per node of the doubling tree

The diagonal families above are the *first cut* of the diagonal vocabulary. A second cut, drawn afterwards, redraws it as twelve elements: YA1–YA6 (yang) and AY1–AY6 (yang mutable), each a set of unit segments along cell diagonals, with integer vertices (axes-v2.json). Write $c = 2e$ for the signed offset of a diagonal line, $x + y = 12 + c$ or $y - x = c$, so that its level is $\cos(\pi c/6)$.

Proposition 3. (i) *The twelve elements are pairwise disjoint, and their union is exactly the set of the 288 cell diagonals of the square; YAk and AYk have $8k$ segments for $k \leq 5$ and 24 for $k = 6$, and the 4095 non-empty unions of elements are 4095 distinct drawings.* (ii) *Every segment of YAk lies on the level $\theta = 180^\circ - 30k^\circ$ and every segment of AYk on $\theta = 30k^\circ$.* (iii) *The elements of one level tile its lines exactly: the level 0° is YA6, the level 180° is AY6, and for $\theta = 30j^\circ$, $1 \leq j \leq 5$, the level is $YA(6-j) \cup AYj$.* (iv) *As band systems $YA(6-j)$ and AYj coincide, so the twelve elements give seven key sets, one per level.* (v) *The lines of AYk are those of YAk shifted by 3 in offset: on the second cut the mutation is $e \mapsto e + 3$, at level $180^\circ - \theta$, as on the orthogonal levels (Proposition 1).*

Proof. Exact check on the segments of axes-v2.json (niveaux_v2.py). ◻

So the second cut is not one more partition of the diagonals: it is one generator for each node of the doubling tree $\theta \mapsto 2\theta$ of §3, read on the diagonal levels (Figure 3). The tree has two towers. Tower A, $90^\circ \rightarrow 180^\circ \rightarrow 0^\circ$, is resolved node by node already by the first cut — $T_1 \text{ YANG MUT}$, $T_0 \text{ YANG MUT}$, $T_0 \text{ YANG}$. Tower B, $\{30^\circ, 150^\circ\} \rightarrow 60^\circ \rightarrow 120^\circ$, is resolved by the first cut only by pairs: $T_2 \text{ YANG}$ ($\equiv T_3 \text{ YANG MUT}$) holds 60° and 120° together, $T_2 \text{ YANG MUT}$ ($\equiv T_3 \text{ YANG}$) holds 30° and 150° together. The second cut separates the two fused pairs, and that is the whole difference.

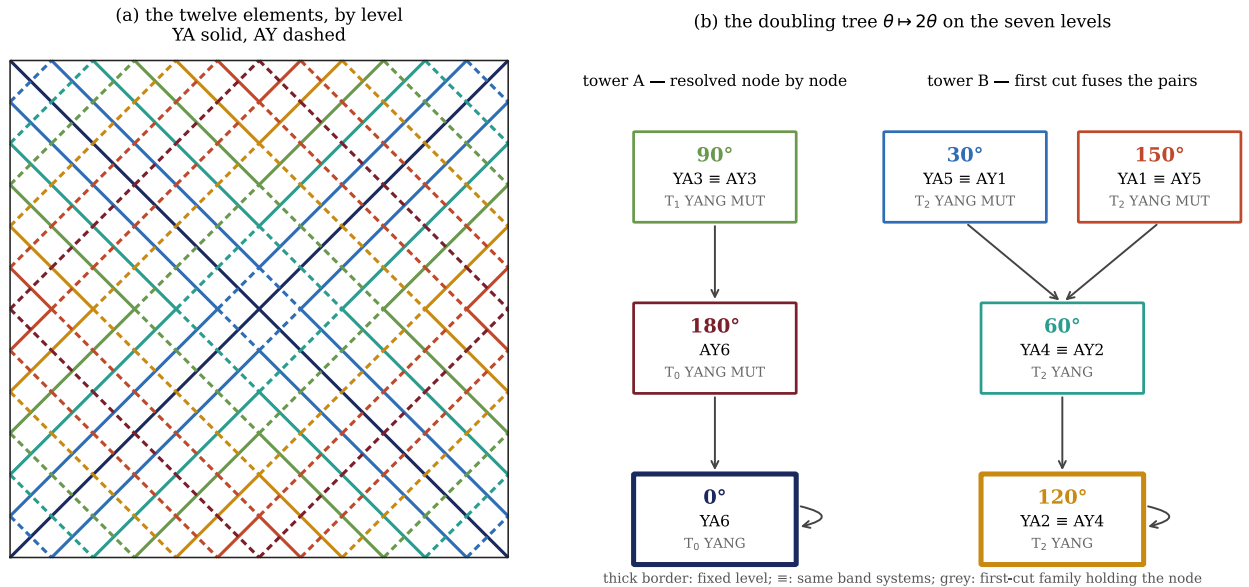


Figure 3. The second cut. (a) The twelve elements on the square, coloured by level, YA solid and AY dashed; the union is the 288 cell diagonals, each once. (b) The doubling tree on the seven levels, with the generator of each node ($\equiv$: same band systems) and, in grey, the first-cut family that holds it. Tower B is where the first cut fuses two nodes into one family.

Theorem 9. *Take as generators the seven diagonal levels of the second cut, $L_0, L_{30}, \dots, L_{180}$, and the eight orthogonal families, unchanged, with the canonical keys and the rule of §5.4.*

- On C_8 they span a code $[1152, 15, 288]$: rank $7 + 8$, no relation, 32 767 non-zero words, and a unique minimum-weight word $L_{180} \oplus T_1$ YIN.
- On C_1 they span $[144, 14, 36]$: rank $7 + 7$, one relation — the eight YIN families sum to zero — 16 383 non-zero words, two minimum-weight words, $L_{180} \oplus T_1$ YIN and $L_0 \oplus L_{30} \oplus \dots \oplus L_{150} \oplus T_1$ YIN MUT, and the all-dark figure is the sum of the seven levels.
- Each code contains the first-cut code of Theorem 6.

Proof. Gaussian elimination and full enumeration of the 2^{15} words (niveauux_v2.py). ◻

Three readings. First, L_{180} is the level of offsets ± 3 , the band systems of T_0 YANG MUT: the minimum-weight word is literally the word of Theorem 6, the diamond against the inner square, on the same 288 triangles. Since the old code is contained in the new and the new adds no lighter word, the distance does not change, and neither does the correction capacity — 287 wrong regions detected, 143 corrected. What grows is the number of figures, not the capacity. Second, the two added dimensions are the two fused nodes of tower B, on C_8 and on C_1 alike: $13 \rightarrow 15$ and $12 \rightarrow 14$. Third, the three relations of Theorem 6 were the price of the fusion; with one generator per node there is nothing to relate on the diagonals, and the only relation left is the orthogonal one at cell level.

A remark on drawings. A level of the second cut is drawn by two elements, which share its segments between them without overlap, but read as band systems the two are one key set. So the parity rule does not see which of the two is drawn: of the 4095 drawings of the second cut, exactly $127 = 2^7 - 1$ are unions of complete level sets, the two elements of a level taken together or not at all. The distinction between the two elements of a level lies below the resolution of the parity; the second cut is a drawing tool as much as a generator set, and the code is taken on the canonical keys.

6. Is the period 12 a choice?

The construction uses the period 12 throughout. Asking whether another period would do is a question about the doubling map alone, and it has an answer.

Read the offsets in $\mathbb{Z}/N$ with the identification $e \sim -e$, since a level is symmetric about the centre.

Theorem 8. *The descent under doubling terminates without a non-trivial cycle if and only if $N = 2^a$ or $N = 3 \cdot 2^a$. A fixed level other than the centre exists if and only if 3 divides N , and it is then the level of offset $N/3$.*

Verified by enumeration for all $N \leq 48$. Without the identification $e \sim -e$ the first statement is false — $N = 12$ then has the cycle $4 \rightarrow -4 \rightarrow 4$ — so the identification is a hypothesis of the theorem and not a convenience.

Consequently a period admits both a terminating descent and a non-trivial fixed level exactly when $N \in \{3 \cdot 2^a\}$, and since the construction needs N even — a half-integer grid, and a diagonal period $N/2$ — that is $\{6, 12, 24, 48, \dots\}$. Write $N = 2^a M$ with M odd; the **depth** of the descent, the number of doublings after which every level is fixed, is a . Depth and number of levels are different quantities and should not be confused: $N = 12$ has seven levels of φ , four of which carry the names $T_0, \dots, T_3$, and depth 2 — the two towers of §3.

Requiring a depth of at least two leaves 12, 24, 48; $N = 6$ closes as well but has depth 1, a single step from every level to a fixed one, and no room for a tower. So 12 is the smallest period that terminates, has a non-trivial fixed level, and has two generations above its roots. For $N = 12$ that fixed level is the offset 4, which is the family T_2 YIN MUT of Theorem 3 and the second root of §2.

Ten, fourteen and eighteen have cycles, and the descent never reaches an end in them. The period is therefore constrained, not chosen freely; what remains a choice is which member of the family 12, 24, 48 one works in.

7. What this is, and what it is not

This is a working basis. We claim no depth for the construction, and we state what it does not settle.

The vocabulary is generated, and the origin is not a choice: it is the set of levels fixed by the doubling, which cannot be reached from anywhere and must be posited, and §6 says which levels those are. What §2.1 adds is that the first generation of one of the two roots reproduces, exactly, a vocabulary fixed beforehand and independently. What is *not* settled here is why the doubling, of all maps on offsets, should be the operation; we take it from the system and derive its consequences.

The counts of §2 are counts of drawn lines in one vocabulary. Changing the origin changes them. They carry no meaning outside the system that defines them, and we do not present them as if they did.

Nothing here is a physical statement. The figures are unions of straight lines on a square; we give no continuous or asymptotic result about them, and nothing in this paper bears on any object outside the finite construction of §1–§5.

8. Related work

Grünbaum and Shephard classified periodic binary fabrics — satins, twills and their relatives — by their symmetry groups, and catalogued the isonemal ones [1,2,3]. The parity rule of §5 is an over-and-under interlacement rule: a region is dark when the number of axes crossed is odd, which is exactly the alternation that makes a fabric hang together. Their classification proceeds from the symmetry group of a given fabric; the construction here proceeds from an origin and one operation, and produces the vocabulary before any figure exists. The two are complementary, and placing the figures of §5 inside their catalogue — which are isonemal, which hang together as real cloth — is an obvious next question that we do not settle.

9. Reproducibility

Every number above is printed by a script that reads the published vector sources and reruns in minutes. Offsets are extracted by exact arithmetic, not by fitting. The extractor asserts that the 1152 regions of a plate are occupied exactly once before any measurement is taken, and refuses files that fail it. The sweeps of Theorems 4 and 5 are over the complete space, not a sample. `niveaux_v2.py` prints Proposition 3, Theorem 9 and the count 127 from `axes-v2.json` and the catalogue, `codes_cles.py` the ranks $5 + 8$ and $5 + 7$, and `fig_v2.py` draws Figure 3.

References

- [1] B. Grünbaum, G. C. Shephard, *Satins and twills: an introduction to the geometry of fabrics*, Mathematics Magazine 53(3) (1980) 139–161.
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